PROJECT PORTFOLIO: INFRASTRUCTURE SCHEDULING & CRITICAL PATH ANALYSIS

Case Study Subject: Dopeni Bridge Construction Project Structural Execution Matrix Academic

Alignment: Project Management (Formative Assessment 1 - Question 4)

Technological Tool Stack: Microsoft Excel & EdrawMax Frameworks

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1. STRUCTURAL PREDECESSOR LOGIC

Project Activity Parameters & Network Logic

- Activity A: Feasibility Study & Site Investigation (No Predecessor) | Duration: 4 Days
- Activity B: Planning & Conceptual Design (Predecessor: A) | Duration: 3 Days
- Activity C: Detailed Engineering Design (Predecessor: A) | Duration: 5 Days
- Activity D: Permits & Approvals (Predecessors: B, C) | Duration: 2 Days
- Activity E: Site Preparation (Predecessor: D) | Duration: 4 Days
- Activity F: Foundation Construction (Predecessor: E) | Duration: 4 Days
- Activity G: Substructure Construction (Predecessor: E) | Duration: 5 Days
- Activity H: Superstructure Construction (Predecessors: F, G) | Duration: 5 Days
- Activity I: Commissioning (Predecessor: H) | Duration: 2 Days

Activity-on-Node (AON) network diagram for the Dopeni Bridge Construction Project, complete with Forward and Backward pass Calculations:

2. ALGORITHMIC FORWARD PASS CALCULATIONS

Calculated using the Day 0 methodology where Earliest Finish (EF) = Earliest Start (ES) + Duration, and ES of

a successor node is defined as max (EF of immediate predecessors).

- Activity A: ES = 0; EF = 0 + 4 = 4
- Activity B: ES = EF(A) = 4; EF = 4 + 3 = 7
- Activity C: ES = EF(A) = 4; EF = 4 + 5 = 9

• Activity D: ES = max (EF(B), EF(C)) = max (7, 9) = 9; EF = 9 + 2 = 11

• Activity E: ES = EF(D) = 11; EF = 11 + 4 = 15

• Activity F: ES = EF(E) = 15; EF = 15 + 4 = 19

• Activity G: ES = EF(E) = 15; EF = 15 + 5 = 20

• Activity H: ES = max (EF(F), EF(G)) = max (19, 20) = 20; EF = 20 + 5 = 25

• Activity I: ES = EF(H) = 25; EF = 25 + 2 = 27

Total Programmed Project Duration Baseline: 27 Days.

3. ALGORITHMIC BACKWARD PASS CALCULATIONS

Calculated sequentially backward from the terminal project duration where Latest Finish (LF) = EF of final node

(27), Latest Start (LS) = LF - Duration, and LF of a predecessor node equals min (LS of immediate successors).

• Activity I: LF = 27; LS = 27 - 2 = 25

• Activity H: LF = LS(I) = 25; LS = 25 - 5 = 20

• Activity G: LF = LS(H) = 20; LS = 20 - 5 = 15

• Activity F: LF = LS(H) = 20; LS = 20 - 4 = 16

• Activity E: LF = min (LS(F), LS(G)) = min (16, 15) = 15; LS = 15 - 4 = 11

• Activity D: LF = LS(E) = 11; LS = 11 - 2 = 9

• Activity C: LF = LS(D) = 9; LS = 9 - 5 = 4

• Activity B: LF = LS(D) = 9; LS = 9 - 3 = 6

• Activity A: LF = min (LS(B), LS(C)) = min (6, 4) = 4; LS = 4 - 4 = 0

4. TOTAL FLOAT MAP & CRITICAL PATH MATRIX

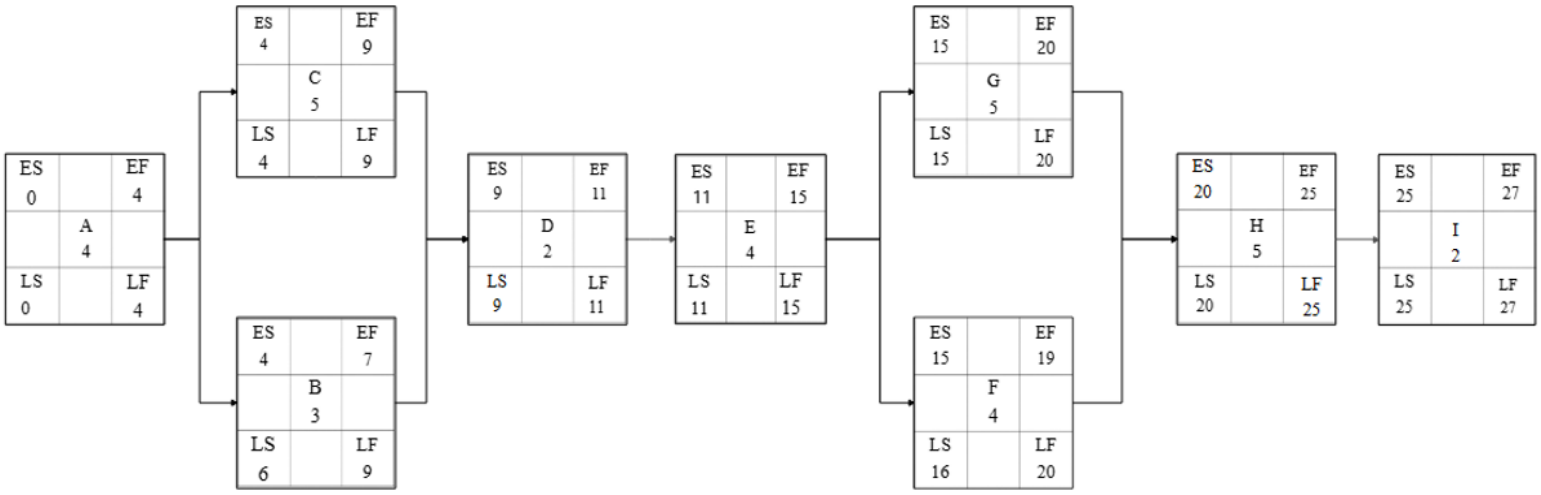
Activity	ES	EF	LS	LF	Total Float	Critical Status
A	0	4	0	4	0	CRITICAL
B	4	7	6	9	2	Non-Critical
C	4	9	4	9	0	CRITICAL

D	9	11	9	11	0	CRITICAL
E	11	15	11	15	0	CRITICAL
F	15	19	16	20	1	Non-Critical
G	15	20	15	20	0	CRITICAL
H	20	25	20	25	0	CRITICAL
I	25	27	25	27	0	CRITICAL

Definitive Project Critical Path Vector: A → C → D → E → G → H → I.

Total Project Duration Requirements: 27 Planned Operational Days.

5. AON LOGIC ROUTING DIAGRAM PREVIEW



6. SCHEDULED GANTT CHART TIMELINE SEQUENCE

Dopeni Bridge Project – Gantt Chart (27 Days):

Activity	Start Day (ES)	Duration (Days)	Project Days
A	0	4	1–4
B	4	3	5–7
C	4	5	5–9
D	9	2	10–11

E	11	4	12–15
F	15	4	16–19
G	15	5	16–20
H	20	5	21–25
I	25	2	26–27

Dopeni Bridge Project – Gantt Chart (27 Days):

Dopeni Bridge Project – Gantt Chart (27 Days)

Activity	Start Day (ES)	Duration (Days)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27
A	0	4																											
B	4	3																											
C	4	5																											
D	9	2																											
E	11	4																											
F	15	4																											
G	15	5																											
H	20	5																											
I	25	2																											

The project duration is 27 days. ES 0 means Activity A begins on project Day 1.